

HALT EXERCISE

X = HALT scores (random variable)

- When things functioning properly

$$\mu = 7 \quad (\text{mean})$$

$$\sigma = 4 \quad (\text{st dev})$$

- Are things functioning properly?

↳ extract a sample $n = 20$ to check

INPUT OF THE PROBLEM

$$\mu = 7 \quad \sigma = 4 \quad (\text{population})$$

$$n = 20 \quad (\text{sample})$$

This is ALL that I need!

↳ From here, you can derive that sample mean $\bar{X}$ has a SAMPLE DISTRIBUTION

$$\bar{X} \sim N\left(7, \frac{4^2}{20}\right)$$

mean of sample mean $\swarrow$ $\searrow$ variance of sample mean

$$SE(\bar{X}) = \frac{4}{\sqrt{20}} = 0.89 \rightarrow \text{st. dev. of sample mean}$$

NB Once I know $\bar{X} \sim N$, I can standardize it to $Z \sim N(0, 1)$: easier computations!

Now, we are ready for any kind of problem on CONTROL LIMITS

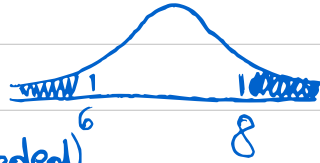
PROBLEM 1: given the CL rule, compute $P_c(\text{Type I error})$

Rule is $6 \leq \bar{X} \leq 8$

$P_c(\text{Type I error}) = ?$

$P_c(\text{taking action when not needed})$

$P_c(\text{tails when } \mu \text{ and } \sigma)$



$$P_c(\bar{X} < 6 \text{ or } \bar{X} > 8)$$

$$= 1 - P_c(6 \leq \bar{X} \leq 8)$$

[now standardize $\bar{X}$ to Z]

$$= 1 - P_c\left(\frac{6-7}{0.89} \leq \frac{\bar{X}-7}{0.89} \leq \frac{8-7}{0.89}\right)$$

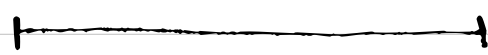
$$= 1 - P_c(-1.1 \leq Z \leq 1.1)$$

you can find this in the z-table

$$= 1 - 0.7287$$

$$= \boxed{0.27}$$

This is $P_c(\text{Type I error})$ given the rule $6 \leq \bar{X} \leq 8$



PROBLEM 2: set $P_c(\text{Type I error})$ find the rule (i.e. CL)

$$P_c(\text{Type I error}) = 0.05$$

You want sth like $P_c(\text{tails}) = 0.05$

From Z-table you know that

$$P_c(Z < -1.96 \text{ or } Z > 1.96) = 0.05$$

Now, express this in terms of $\bar{X}$ (rather than Z)

we know

$$\frac{\bar{X} - 7}{0.89} = Z \sim N(0, 1)$$

Hence,

$$Z = \frac{\bar{X} - 7}{0.89} < -1.96 \Rightarrow \bar{X} < 7 - 1.96(0.89) \approx \boxed{5.26}$$

$$Z = \frac{\bar{X} - 7}{0.89} > 1.96 \Rightarrow \bar{X} > 7 + 1.96(0.89) \approx \boxed{8.74}$$

In order to have $P_c(\text{Type I error}) = 0.05$, Control Limit should be set to $[5.26 \leq \bar{X} \leq 8.74]$